IIOT NO 4 ROLL NO 56 B

```
#include <WiFi.h>
#include <HTTPClient.h>
const char* ssid = "Wokwi-GUEST";
const char* password = "";
String apiKey = "09R78TVR7UBXP6J5";
const int potentiometerPin = 33;
const char* server = "http://api.thingspeak.com";
void setup() {
  Serial.begin(115200);
  WiFi.begin(ssid, password);
  Serial.print("Connecting to Wi-Fi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.print(".");
  }
  Serial.println("\nConnected to Wi-Fi");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}
void loop() {
  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;
    int potValue = analogRead(potentiometerPin);
    float voltage = (potValue / 4095.0) * 3.3;
    Serial.print("Potentiometer Value: ");
    Serial.println(potValue);
    Serial.print("Voltage: ");
    Serial.println(voltage);
    String url = String(server);
    url += "/update?api_key=" + apiKey;
    url += "&field1=" + String(potValue);
    url += "&field2=" + String(voltage, 2);
    http.begin(url);
    int httpResponseCode = http.GET();
    Serial.print("HTTP Response Code: ");
```

```
  Serial.println(httpResponseCode);
  if (httpResponseCode > 0) {
   Serial.println("Response: " + http.getString());
  }
  http.end();
 } else {
  Serial.println("Wi-Fi Disconnected");
 }
 delay(20000);
}
```



